RESEARCH IMPLEMENTATION RECORD

Vehicle Tracking + Occlusions Research Project

Phase 1A, Phase 1B and Phase 1C
Literature-guided tracker planning, evaluation infrastructure and comparative baseline experiments

Purpose of this document

A precise record of what was done, why it was done, how it was implemented, what the experiments showed, and which findings are still provisional. Written to support the subsequent methodology, experimental design and final manuscript.

RECORD	STATE AS OF 04 OCTOBER 2026
Research direction	Working topic fixed; exact recovery policy and novelty claim not experimentally established
Phase 0	Previously frozen; its original records are preserved
Phase 1A	Four tracking/occlusion papers reviewed for module design
Phase 1B	Event loader, tracking interface, runner and evaluation skeleton implemented
Phase 1C	SAM2.1 Tiny and DAM4SAM Tiny run on all 12 pilot events; consolidation and audits through Step 7.3 completed
Not yet done	Step 7.4 decisions, Phase 1C formal freeze (Step 8), later Phase 1 stages, proposed sequential cue-acquisition experiments

Source basis. Finalized Working Research Proposal v2 (September 2026), Phase 0 Frozen Documentation (28 September 2026), and the Phase 1 commands, output tables, manual frame reviews and audit CSV contents provided during the project. No new experiments or external literature verification are claimed.

Document map and evidence conventions

This is a Phase 1 record, not a rewrite of the proposal or Phase 0. The supplied documents define the research question, scope and original event inventory; this record describes subsequent implementation, testing and corrections. The research-topic PDF retains authority for proposed future modules; the experiment and audit data are limited to what has actually been run.

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Evidence labels used throughout

LABEL	MEANING
Measured	Output from a pasted JSON/CSV or terminal evaluation, using the project evaluator.
Visually reviewed	Observation from inspected overlay images, including user-reported checks where upload limits prevented independent image review.
Provisional	A scenario boundary or inferred recovery/identity mechanism that requires stronger visibility-aware review.
Planned	A research-system component or experiment described in the proposal but not built/tested in Phase 1A-1C.

Reading rule

A successful XML-box match is not, on its own, post-occlusion identity recovery. During these tests a target may remain annotated while fully hidden, a tracking box may persist without physical visibility, and a possible-switch warning is only a heuristic flag. The manual overlays qualify these outputs.

1. Research goal and connection to Phase 1

Working research title: Reliability-Aware Sequential Identity Evidence Acquisition for Post-Occlusion Recovery of User-Selected Road Vehicles.

Frozen primary research question

Can reliability-aware sequential acquisition of vehicle identity cues improve post-occlusion recovery of a user-selected road motor vehicle among visually similar distractors, while reducing false identity decisions and average inference cost compared with always-on multimodal Re-ID and fixed coarse-to-fine recovery?

The system concerns one explicitly selected road motor vehicle (cars/SUVs/vans, motorcycles, buses and trucks). The future recovery mechanism considers vehicle color, body type, plate, make/model, learned visual Re-ID, local details and temporal plausibility, with explicit cue reliability, availability, evidence value and acquisition cost. Candidate decisions are MATCH, REJECT or NEED MORE EVIDENCE. Those are proposed later components, not Phase 1C implementations.

Why Phase 1 came first: the project needed a functional user-selected tracker and a consistent evaluator before it could test whether sequential identity evidence improves recovery. Otherwise, errors from annotation parsing, initialization, mask propagation or inaccurate event labels could be mistaken for failures of the proposed method.

The research proposal explicitly warns that coarse-to-fine Re-ID, distractor-aware memory, multimodal plate/appearance fusion and generic cheap-to-expensive cascades are prior art. Consequently, these baseline trackers are comparators and enabling infrastructure; no Phase 1 result demonstrates the final novelty claim.

2. Phase 0 handoff: the pilot event dataset

Phase 0 supplied a frozen set of 12 annotated pilot events E001-E012 extracted from UA-DETRAC XML and manually checked in original frames. It used three test-annotation sequences: MVI_39311 (1,505 frames), MVI_39401 (1,385 frames) and MVI_40761 (2,030 frames). BDD100K had been acquired as a possible later gap-filling source but was not the source of the frozen Phase 1C tests.

Phase 0 deliberately separates the XML-described occlusion intervals, target IDs, bounding boxes and candidate reappearance frames from manual judgments about what the target actually shows. Its frozen event and coverage CSVs are inputs to Phase 1 rather than datasets to re-label silently.

INHERITED ASSET	PHASE 1 USE
data/processed/pilot_events/pilot_event_log.csv	Frozen event IDs, sequences, target IDs, event anchors and original scenario notes.
data/processed/pilot_events/scenario_coverage.csv	Initial mapping of 10 stress scenarios (S01-S10) to pilot events.
data/raw/UA-DETRAC/images/<sequence>/img#####.jpg	Consecutive video frames consumed by each single-target tracking run.
data/raw/UA-DETRAC/annotations/test_xml/.../<sequence>.xml	Selected-target boxes and other vehicle annotations for initialization and evaluation.
Phase 0 contact sheets and observation notes	Visual confirmation of target identity and plausible occlusion/reappearance intervals.

Protocol consequence: these three repeatedly viewed test-annotation sequences are development/pilot material. Their repeated use is documented; a future, untouched test set is necessary for credible final evaluation.

3. Phase 1A - Literature-guided tracker and recovery design

Status: Phase 1A functional reading/review completed for four works: VeTrack, Sentinel, DAM4SAM and DeconfuseTrack. The wider literature program included vehicle identity, sequential feature acquisition and reject-option papers, but the implementation-focused Phase 1A set was this tracking/occlusion group. The purpose was to map prior solutions onto specific modules, not prematurely select the final evidence-acquisition algorithm.

WORK	RELEVANCE EXTRACTED FOR THIS PROJECT	USE / BOUNDARY
VeTrack	Vehicle appearance/motion information and confidence-guided multi-level association under difficult traffic tracking.	Suggests confidence-conditional association; routine association is existing work, not our novelty.
Sentinel	Per-track confidence/uncertainty and explicit reliability/loss reasoning under extended occlusion.	Supports a future tracking-reliability/loss trigger; no loss-state algorithm adopted yet.

WORK	RELEVANCE EXTRACTED FOR THIS PROJECT	USE / BOUNDARY
DAM4SAM	Separates ordinary appearance updating from distractor-aware memory for SAM2.1.	Selected as an existing distractor-aware baseline, not invented or newly trained by this project.
DeconfuseTrack	Ambiguity/deconfusion and potentially occluded competing candidates during object tracking.	Motivates ambiguity and distractor tests; not claimed to have been implemented as a separate tracker.

Phase 1A output and decision

The review justified a simple, strong baseline comparison: T0 = standard SAM2.1 and T1 = DAM4SAM. It reinforced the need to distinguish occlusion from genuine loss, use cautious memory updates, and measure distractor errors. It did not finalize the proposed sequential policy, cue reliability equations, candidate generator or vehicle-Re-ID fusion.

4. Phase 1B - Common execution and evaluation framework

4.1 Implemented project modules

PATH / COMPONENT	ACTUAL ROLE
src/evaluation/event_loader.py	Reads a frozen event and UA-DETRAC XML; derives selection frame and target xyxy box; exposes event boundaries, sequence and target metadata.
src/tracking/base_tracker.py	A common tracker contract so alternative trackers can return compatible predictions.
src/tracking/debug_static_tracker.py	Static-box sanity checker; exercises the whole pipeline without meaningful visual tracking.
src/run_phase1_event.py	Loads frames, initializes selected target, advances tracker and saves predictions and run metadata.
src/evaluation/event_evaluator.py	Computes XML-box matching and possible-switch diagnostics; writes framewise and event-level results.
src/evaluation/metrics.py	Evaluation helpers and bookkeeping used by the event evaluator.
src/{tracking,evaluation,recovery}/	Organized Phase 1 source directories; recovery directory does not itself mean a recovery system was implemented.
configs/phase1/, results/phase1/	Configuration and baseline output locations.

4.2 Event contract and initialization

Each event specifies sequence, selected target ID, selection frame/box, occlusion_start, optional maximum_occlusion_start, optional reappearance_frame, and an evaluation end frame. The selection box is in xyxy pixel coordinates. The tracker is initialized only once on the selected vehicle; it must then propagate that identity through subsequent frames.

The original initialization heuristic looked backward up to 30 frames before the recorded occlusion and preferred a target annotation with no occlusion tag. This worked in many events but failed in some legitimate pilot cases. The following bounded, auditable exceptions were introduced:

EVENTS	EXCEPTION	REASON
E003, E005, E009	Extend candidate selection lookback to 120 frames.	Clean candidate earlier than 30 frames; verified selection frames 895, 776 and 945.
E001	Manually use frame 50 and the XML target box.	All target-5 frames 1-50 had an occlusion tag, yet the selected vehicle was visibly identifiable in frame 50.
Other events	Retain the ordinary selection behavior.	Avoid introducing global changes for isolated annotation/visibility conditions.

E001 initially triggered a failed preflight and was deferred until this override passed compilation, dry run and visual inspection. For E009, an XML check found 120 annotated target frames in the examined lookback interval and only 23 unoccluded frames, ending at frame 945. These fixes concern initialization eligibility, not changes to the frozen Phase 0 event definitions.

4.3 Output/evaluator meaning

For every future frame, the common prediction record includes target presence, predicted region (xyxy, where available), and tracker runtime. The evaluator compares the selected target against the same-ID UA-DETRAC XML box using a bounding-box IoU threshold of 0.30 for a match. A possible-switch flag warns of an overlap or identity issue but is not a verified ID-switch count. A preliminary recovery-frame heuristic requires a short consecutive run of matches after the recorded reappearance anchor (three matched frames in the implementation).

Outputs include predictions.csv, run_metadata.json, evaluation_frames.csv and evaluation_summary.json for each tracker-event run. Recorded recovery delay can be zero even if the tracker never became lost; this is a limitation of the current evaluator, not evidence of instant reacquisition.

4.4 Phase 1B sanity test: E011 static tracker

The debug runner initialized E011 at frame 309, evaluated frames 310-408 (99 future frames) and successfully executed the file/CLI path. It reported 7 target-matched frames, 10 possible-switch flags (first at frame 317), and no recovery frame. This result is only a workflow sanity check: a constant bounding box is deliberately not a valid visual tracking baseline.

5. Phase 1C - Baseline selection, setup and execution

5.1 What exactly was compared

BASELINE	IMPLEMENTATION	ROLE
T0 - SAM2.1 Tiny	Standard SAM 2.1 Hiera-Tiny video propagation, user-selected initial region.	Reference tracker-only/streaming-memory behavior.
T1 - DAM4SAM Tiny	Authors' distractor-aware SAM2.1-based tracker in tiny configuration.	Existing distractor-aware tracking/memory comparator.

Both used the same manually validated event identity and initialization conditions within each event. The project did not train either model from scratch or implement plate OCR, vehicle Re-ID candidate retrieval, calibrated reject decisions or sequential cue selection in this phase.

5.2 Implementation environment and reproducibility

ITEM	RECORDED CONFIGURATION
Operating system / project	Ubuntu Linux; local project root: /home/alpha/Desktop/Paper/Vehicle Tracking Paper
GPU	NVIDIA Quadro P2000; 4 GB VRAM; compute capability 6.1
SAM2 Conda environment	vehicle_sam21; Python 3.10; PyTorch 2.5.1 + CUDA 12.1
DAM4SAM Conda environment	vehicle_dam4sam; Python 3.10; PyTorch 2.1.0 + CUDA 12.1; NumPy 1.26.4
Research source checkouts	external/sam2/ and external/DAM4SAM/ (external codebases kept outside project-authored implementation)
Checkpoint / configuration	SAM 2.1 Hiera Tiny checkpoint: external/sam2/checkpoints/sam2.1_hiera_tiny.pt; DAM4SAM referenced the same checkpoint via symlink; corresponding tiny YAML configurations
Installation note	Optional SAM2 CUDA post-processing extension not built; warning acknowledged. This does not establish that outputs are mathematically identical to a fully built installation.
Run level	One selected vehicle per event, continuous sequence of original video frames, no reinitialization from GT at recorded reappearance.

Running times observed here were roughly 1.0-1.13 seconds per frame on this machine, so these pilot configurations were not real-time in this environment. Runtime is tracker time as logged by the runner, not cue-acquisition cost or end-to-end system energy.

5.3 Five-step per-event workflow

STEP	ACTION	WHY IT WAS NEEDED
1. Preflight	bash scripts/run_phase1c_event.sh E00X --dry-run; confirm selection preview.	Prevent wrong-target initialization and invalid event ranges.
2. T0 execution	Run the selected interval with SAM2.1 Tiny.	Obtain unmodified tracker-only prediction trajectory.
3. T1 execution	Run the exact event with DAM4SAM Tiny.	Test distractor-aware memory under same event identity.
4. Evaluation	Write framewise/summary metrics and comparison_summary.json.	Maintain a common numerical diagnostic protocol.
5. Visual audit	Review key frames around occlusion, reappearance, ambiguity and event end.	Distinguish actual vehicle identity/visibility from XML overlap and heuristic flags.

The event-by-event Bash wrapper, scripts/run_phase1c_event.sh, was introduced to automate the repeated steps without batching all remaining events into one opaque run. Thus, the researcher could inspect and correct each event before moving to the next.

6. Exactly which intervals were tested

The following table records the event selection that was actually used in Phase 1C. A recorded Phase 0 reappearance anchor is a review point, not a demonstrated loss/reacquisition boundary.

EVENT	SEQ / ID	INIT	TESTED FRAMES	N	SCENE NOTES
E001	39311 / 5	50	51-202	152	Partial occlusion; manual initialization
E002	39311 / 5	199	200-356	157	Severe partial; bus
E003	39401 / 92	895	896-1046	151	Provisional bus full occlusion

EVENT	SEQ / ID	INIT	TESTED FRAMES	N	SCENE NOTES
E004	39401 / 75	633	634-750	117	Pole/traffic; boundary
E005	40761 / 4	776	777-1323	547	Stationary SUV; two long bus occlusions
E006	40761 / 4	1070	1071-1279	209	Second SUV occlusion interval
E007	40761 / 4	1321	1322-1443	122	Later SUV occlusion interval
E008	39401 / 39	242	243-397	155	Moving SUV; roadside obstruction
E009	39311 / 35	945	946-1143	198	Moving taxi; 120-frame lookback
E010	39311 / 35	1376	1377-1505	129	Originally called target absent; corrected
E011	40761 / 24	309	310-408	99	Moving taxi, similar distractors
E012	40761 / 24	461	462-608	147	Repeated occlusion/near-exit ambiguity

A total of 2,183 event-frame positions per tracker were processed across the 12 event windows (4,366 tracker-event frames for both trackers together). This is not 2,183 independent frame observations: overlapping windows and repeated sequences/vehicles are present.

7. Exact baseline metrics as consolidated in Step 7.1

Measured source: results/phase1/phase1c_baseline_metrics.csv, generated from the per-event evaluation summaries and run metadata. “Match” means XML-based target IoU >= 0.30, and “flags” are only possible-switch warnings. All times are average tracker runtime in milliseconds/frame.

EVENT	EVAL	XML GT	SAM MATCH	DAM MATCH	SAM FLAGS	DAM FLAGS	SAM ms	DAM ms
E001	152	152	152	152	0	0	1080.8	1126.0
E002	157	157	157	157	0	0	1084.6	1127.6
E003	151	141	141	141	0	0	1105.8	1111.9
E004	117	117	116	116	0	0	1076.2	1103.4
E005	547	547	223	226	0	0	1097.3	1128.1
E006	209	209	63	62	24	0	1083.5	1087.3
E007	122	122	57	57	0	0	1084.0	1018.3
E008	155	155	155	155	0	0	1081.9	1129.9
E009	198	198	198	198	0	0	1086.1	1134.3
E010	129	129	129	129	0	0	1077.2	1124.4
E011	99	99	99	99	0	0	1082.3	1100.3
E012	147	140	136	139	18	2	1082.3	1119.9

Unweighted mean tracker runtime across the 12 event means: SAM2.1 1,085.17 ms/frame; DAM4SAM 1,109.28 ms/frame. Event-frame-weighted means: 1,087.65 and 1,114.08 ms/frame, respectively. SAM2.1 was numerically faster in 11/12 individual runs; E007 favored DAM4SAM. These comparisons reflect a single local machine/configuration, not a comprehensive speed benchmark.

Do not report these as overall recovery accuracies

Large reductions in XML match counts for E005-E007 primarily coincide with long annotated occlusions and prediction abstentions, so ratios such as 223/547 cannot be used as direct recovery success rates. Similarly, an evaluator-recorded zero recovery delay for a never-lost tracker is not a recovered-after-loss event.

8. Event-by-event visual findings and corrected interpretation

The following narrative reports the frame evidence inspected during the research conversation. It explicitly separates successful tracking continuity, plausible resumption following absence, and ambiguous/incorrect predictions.

E001 | Partial-occlusion initialization, target 5

MVI_39311; init frame 50 by manual override; bus-related occlusion begins 51; recorded reappearance 172. Both trackers placed correct boxes at 51-52, 171-173 and 202. Only the blue GT box at initialization frame 50 was expected because predictions start at frame 51.

Interpretation: Continuous tracking from partially occluded initialization; no tracker loss or post-loss recovery demonstrated.

E002 | Severe partial occlusion, target 5

MVI_39311; init 199; reference frames 200-201, 224-226, 270-272 and 356. Both green boxes remained aligned with the selected vehicle as a bus obscured part of it. No possible-switch flags, 157/157 XML matches for each tracker.

Interpretation: Successful robustness under partial evidence; bus cover does not prove complete disappearance.

E003 | Provisional full-occlusion candidate, target 92

MVI_39401; init 895 after extending lookback; bus interaction around 955/976 and recorded reappearance 1016. Both trackers remained present through the sampled interval and aligned after 1016. Counts: 141/141 annotated matches for each across 151 processed future frames.

Interpretation: The Phase 0 full-disappearance label remains provisional; imagery reviewed in Phase 1C appeared compatible with continuing partial visibility. No confirmed reacquisition after tracker loss.

E004 | Partial occlusion, pole and eventual boundary, target 75

MVI_39401; init 633; reference frames 634-635, 674-676, 719-721 and 750. Both trackers followed the target, including movement toward the left edge. Each had 116/117 XML matches; no switch flags.

Interpretation: Good continuity and boundary behavior; one unmatched frame per tracker does not independently establish identity loss.

E005 | Long repeated occlusions of a stationary SUV, target 4

MVI_40761; init 776; explicit run through 1323 to include two bus occlusions. SAM2.1 absent approximately frames 882-1043 and 1094-1237; later boxes aligned with the parked SUV near 1063 and 1249. Early narrow boxes near 1044 and 1238 could include bus-edge pixels. Matched XML frames: SAM 223, DAM 226 of 547 annotations.

Interpretation: Visible tracking resumed after prolonged prediction absence for a stationary target. Exact first reliable identity recovery and visibility boundary not independently fixed.

E006 | Stationary SUV, false continuation stress

MVI_40761; init 1070; reference frames 1083-1110 and 1249/1279. SAM2.1 produced small green predictions on unrelated/non-target visual regions around 1107-1110 while an occluding bus passed; DAM4SAM more often withheld prediction. The evaluator recorded 24 SAM possible-switch flags versus 0 DAM flags, with 63 versus 62 XML matches.

Interpretation: Important possible false continuation behavior, favoring DAM4SAM qualitatively on those frames; do not call all 24 flags confirmed ID switches. Same stationary target as E005/E007.

E007 | Later stationary SUV occlusion

MVI_40761; init 1321; both trackers absent during part of long occlusion. Narrow uncertain boxes appeared around frame 1395 (SAM) and 1396 (DAM). Both later aligned with the original stationary SUV at 1412-1414 and 1443. Each recorded 57/122 XML matches and 0 flags.

Interpretation: Both showed apparent resumption after absence; earliest narrow predictions should not be counted as established identity recovery.

E008 | Moving SUV among distractors, target 39

MVI_39401; init 242; both maintained the dark SUV through roadside pole/sign obstruction at 285-287, around recorded reappearance 350, and at 397. Both produced predictions in all 155 frames and 155 XML matches with 0 switch flags.

Interpretation: Useful moving-target control case with multiple vehicles, but tracker presence never dropped; not a genuine post-loss challenge.

E009 | Moving partial occlusion, target 35

MVI_39311; no clean frame in 30-frame lookback before frame 1043; 120-frame lookback selected frame 945. Both continued predictions at 1042-1044, 1074-1076 and 1112-1113. For SAM2.1 at 1114 and 1143, the user reported roughly 90% and 98% visual box overlap (not computed IoU). Both 198/198 XML matches, 0 flags.

Interpretation: Continued tracking; the two final overlap percentages are user estimates, not independently measured IoU.

E010 | Original target-absence label contradicted

MVI_39311; init 1376, target taxi 35; frames 1390-1392 remained visibly on the original taxi. At frame 1505 a car obscured most of it, but the roof and left headlight remained visibly present. Both trackers predicted continuously and matched all 129 XML boxes.

Interpretation: This event does not satisfy strict target absence. A prior no-reappearance-to-end label was insufficient; S07 is not covered by this event.

E011 | Moving taxi among visually similar taxis, target 24

MVI_40761; init 309; overlap with similar-colored taxis in frames around 353-380; both trackers remained aligned with target through frame 408. Both logged 99/99 matches, 0 flags, no absence.

Interpretation: Valid similar-distractor control; no fully lost moving-target identity recovery demonstrated.

E012 | Repeated moving-taxi occlusion and scene exit

MVI_40761; init 461; from frame 462 to 608. There were 140 frames annotated for target 24 (through 601), then no same-ID target annotation 602-608. SAM had 136/140 matches and 18 possible-switch flags, with apparent drift toward target-ID-2 region and continued presence past annotated exit. DAM had 139/140 matches, 2 flags, an early absent prediction at 586, and mostly stopped after frame 603.

Interpretation: Most informative moving-case ambiguity/false-continuation comparison. “Fully visible at 608” in one original annotation/log field must not be interpreted as reappearance: manual review instead described the target exiting around that boundary.

9. Consolidated results - Step 7.1 and Step 7.2

9.1 Step 7.1 - Machine-readable metric consolidation

A one-off Python consolidation over the 12 event folders read evaluation_summary.json and run_metadata.json for both trackers. It created results/phase1/phase1c_baseline_metrics.csv with 24 rows (12 events x 2 trackers): evaluated frames, annotated frames, XML matches, possible-switch flags and mean tracker runtime. All 24 records were present.

9.2 Step 7.2 - Qualitative/failure audit

A second script produced results/phase1/phase1c_failure_audit.csv, retaining the original event identities and adding scenario_review, observed_behavior, failure_assessment, per-tracker matches/flags, overlapping_video indicator, review_notes and audit_status=preliminary. It was reviewed against the visual evidence.

OUTCOME CATEGORY	EVENTS	COUNT	INTERPRETATION
Tracking continuity	E001, E002, E003, E004, E008, E009, E010, E011	8	No observed loss in the recorded tracker outputs; includes a provisional full-occlusion label (E003).

OUTCOME CATEGORY	EVENTS	COUNT	INTERPRETATION
Prediction loss then visual resumption	E005, E006, E007	3	All use the same stationary SUV; cannot establish moving-target recovery capability.
Ambiguity near target exit	E012	1	Comparison of continued incorrect/uncertain tracking versus abstention.

The audit treated E006 and E012 as the strongest qualitative error-analysis cases, E010 as a scenario-classification correction, and E005-E007 as repeated but correlated stationary-target episodes. All statuses remained preliminary because current metrics are not visibility- or identity-decision-aware.

9.3 Comparative interpretation without overstating superiority

• SAM2.1 vs DAM4SAM in easy/continuous cases: performance was closely matched by the chosen XML IoU criterion; no gain in recovery ability is inferable from 100% matched annotations.

• DAM4SAM and ambiguity: E006 had 0 vs 24 possible-switch flags, and E012 had 2 vs 18; visual examples in these events support more cautious behavior by DAM4SAM. These do not form a full event-level identity-switch-rate estimate.

• Computation: SAM2.1 was faster in 11 of 12 event-local runtime averages. The only measured cost here is tracker runtime, so the claimed benefits of selective identity cue acquisition remain untested.

• What the evaluation is currently missing: verified loss onset, reappearance with visibility state, candidate recall, correct/false/rejected recovery decisions, and cost of extra cues. These are later evaluation requirements, not Phase 1C results.

10. Phase 1C Step 7.3 - Scenario coverage correction

Using the original 10 scenario definitions, a script read the Phase 0 scenario coverage CSV and generated results/phase1/phase1c_scenario_audit.csv with original status, reviewed status, supporting events and evidence notes. The frozen Phase 0 source was not overwritten.

ID	SCENARIO	PHASE 1C STATUS	SUPPORT	EVIDENCE BOUNDARY
S01	Short occlusion then reappearance	Covered	E001, E002, E011	Partial occlusion with continued tracking established.
S02	Same-color/body-type distractors	Covered	E011, E012	Similar taxis appear; no strong post-loss recovery proof.
S03	Nearly identical vehicles together	Provisional	E011, E012	Near-identity appearance similarity not formally tested.
S04	Unreadable/unavailable plate	Provisional	E003, E011, E012	Actual plate quality/cue availability not measured.
S05	Changed viewpoint	Provisional	E008, E012	Strong viewpoint-change recovery not isolated.
S06	Long disappearance	Covered - limited	E005, E006	Prolonged occlusion, stationary SUV only.
S07	Target absent during observation window	Missing	None	E010 remained partially visible at final frame.
S08	Distractors appear before target	Missing	None	No verified distractor-before-target reappearance case.
S09	Repeated occlusions	Covered - limited	E005, E006, E007	Repeated bus occlusions of same stationary vehicle.

ID	SCENARIO	PHASE 1C STATUS	SUPPORT	EVIDENCE BOUNDARY
S10	Lighting/resolution changes	Provisional	None verified	Mixed day/night videos but no validated within-event transition.

Coverage total: 2 covered (S01/S02), 2 covered with limitations (S06/S09), 4 provisional (S03/S04/S05/S10), and 2 missing (S07/S08). These are scenario readiness labels, not 20%/40% accuracy measures.

Source-version inconsistency to preserve

The uploaded Phase 0 frozen documentation already treats strict S07 as missing (its Section 8 notes that E010 does not establish physical target absence), but the scenario_coverage.csv text used during Phase 1C was shown with S07 marked covered by E010. The Phase 1C audit retained that incoming CSV status in phase0_status and corrected phase1c_status to missing. Reconcile these file versions before preparing any final dataset appendix; do not alter the original frozen artifacts retroactively.

11. Threats to validity and safeguards identified

11.1 Visibility is not guaranteed by the XML

UA-DETRAC boxes and occlusion metadata can persist through a target's visual disappearance. Treating every GT-annotated frame as an equally observable positive example would make a useful abstention look like failure and a stale prediction look like success. Further, preliminary ground-truth reappearance anchors may lag the first visible pixels or capture boundary exit rather than genuine return.

11.2 Tracking presence is different from identity confidence

The tracker adapter returns a presence/box decision derived from the propagated mask. This is not a calibrated accept/reject decision over all road-vehicle candidates. Early narrow strips at occluder boundaries are particularly uncertain; green boxes are not evidence of verified identity without visible target cues.

11.3 Sample dependence

E001-E002 share vehicle 5; E005-E007 share a parked SUV and contain overlapping/adjacent observation periods; E009-E010 share vehicle 35; E011-E012 share moving taxi 24; all events come from only three repeatedly inspected recordings. Per-frame pooling or treating all 12 as independent scenes would overstate generalizability. A future benchmark must separate development and untouched evaluation recordings.

11.4 Scope of the baseline conclusion

The pilot supports the operational ability to initialize and propagate a manually selected vehicle, to record per-event outputs, and to identify situations of misleading tracker continuation. It does not quantify full moving-target recovery accuracy among near-identical distractors, final false-match/reject performance, candidate generation recall, reliability calibration or sequential computation savings.

12. Research artifacts and how to reproduce this stage

12.1 Directories and key outputs

TYPE	PATH
Frozen input	data/processed/pilot_events/pilot_event_log.csv data/processed/pilot_events/scenario_coverage.csv
Annotations	data/raw/UA-DETRAC/annotations/test_xml/DETRAC-Test-Annotations-XML/MVI_*.xml
Images	data/raw/UA-DETRAC/images/MVI_39311, MVI_39401, MVI_40761
Baseline adapters	src/tracking/sam21_tracker.py src/tracking/dam4sam_tracker.py
Runner/evaluation	src/run_phase1_event.py src/evaluation/event_loader.py src/evaluation/event_evaluator.py
Orchestration	scripts/run_phase1c_event.sh
Per-run output	results/phase1/<event>/<tracker>/predictions.csv results/phase1/<event>/<tracker>/run_metadata.json results/phase1/<event>/<tracker>/evaluation_frames.csv results/phase1/<event>/<tracker>/evaluation_summary.json
Event comparison	results/phase1/<event>/comparison_summary.json and visual-review images
Aggregate evidence	results/phase1/phase1c_baseline_metrics.csv results/phase1/phase1c_failure_audit.csv results/phase1/phase1c_scenario_audit.csv

12.2 Representative commands already used

```
python -m py_compile src/evaluation/event_loader.py
bash scripts/run_phase1c_event.sh E001 --dry-run
bash scripts/run_phase1c_event.sh E001
cat results/phase1/E001/comparison_summary.json
python src/run_phase1_event.py --event E011 --tracker debug
python src/evaluation/event_evaluator.py --event E011 --tracker debug
```

These are representative reproducing commands, not a claim that external repositories, environment installations or model checkpoints are packaged with this PDF. The precise repository revision identifiers, random seeds and full dependency lockfiles were not supplied in the available Phase 1 conversation; do not fabricate them when writing a reproducibility statement.

13. Status and handoff to subsequent research work

MILESTONE	STATUS	DOCUMENTED RESULT
Phase 1A	Completed	Functional study of VeTrack, Sentinel, DAM4SAM and DeconfuseTrack; selected candidate baselines and deferred novel methodology choices.
Phase 1B	Completed	Single-target event loader, tracker interface, debug path, common runner and initial evaluation outputs.
Phase 1C Step 6	Completed	SAM2.1 Tiny and DAM4SAM Tiny across E001-E012: 24 tracker-event runs, reviewed eventwise.
Phase 1C Step 7.1	Completed	Consolidated baseline metrics CSV (24 rows).
Phase 1C Step 7.2	Completed / preliminary	Failure audit CSV (12 rows), visibility and independence caveats preserved.
Phase 1C Step 7.3	Completed / preliminary	Scenario coverage audit CSV (10 scenarios) with strict S07/S08 gaps.
Phase 1C Step 7.4	Not performed	Priority/readiness decision was discussed but no executed step or formal new-event selection recorded.
Phase 1C Step 8	Not performed	No formal freeze report/acceptance for Phase 1C yet.
Later phases / proposed method	Not performed	Candidate generation, loss detection, actual MATCH/REJECT/MORE and sequential evidence policy still to implement/evaluate.

Phase 1 handoff decision, not a completed experiment

Proceed using this work as a baseline pilot and error-analysis record. Prioritize a small number of genuinely difficult moving-vehicle full-loss/reappearance cases, strict target-absent negative cases (S07), and distractor-before-target cases (S08) before attempting final claims. A single event may cover more than one requirement. Avoid rerunning unchanged easy events without a specific methodological need.

Publication-ready takeaway at this stage: On this 12-event UA-DETRAC pilot, two established SAM2.1-based baselines often maintained identity during partial occlusion. Selected heavy-occlusion events showed prediction absence followed by visual tracking resumption for a stationary SUV, while E006/E012 exposed possible false continuation and distractor ambiguity. The current protocol is insufficient to claim quantitative post-loss moving-vehicle recovery superiority or computational savings from sequential identity evidence. This is a valid research-stage result and directly determines what the subsequent system and evaluation must add.

Appendix A. Numerical verification and interpretation guide

TERM / VALUE	PRECISE INTERPRETATION
evaluated_frames	Future frames processed by evaluator, not necessarily visible target frames.
target_annotated_frames	Frames with same-ID XML annotation; can include severe/full visual occlusion.
target_matched_frames	IoU >= 0.30 for prediction versus annotated target box, not verified recognition.
possible_switch_frames	Diagnostic warnings; not confirmed ground-truth ID switches.
prediction_absent_frames	Frames the tracker adapter did not output a present target; not automatically a correct reject.
recovery_frame / delay	Early heuristic of matched-box sequence after logged reappearance; unreliable as post-loss delay if never lost.
runtime_ms	Average local tracker execution time per future frame, not OCR/Re-ID/fusion cost.
visually checked correct	Per-frame overlay observations, with uncertainty where object barely visible or only user estimates are available.

Appendix B. Critical continuity notes

• E001: explicit manual selection at frame 50; this is a special partially occluded initialization case. Keep it distinct from clean-initialization events.

• E003: Phase 0 candidate for full occlusion; Phase 1 observations did not certify complete unseen interval. Do not relabel as confirmed recovery.

• E005-E007: strong long/repeated occlusions but same parked SUV. Some earliest green predictions were likely occluder-edge fragments.

• E006: 24 possible-switch flags for SAM2.1 are not 24 verified identity switches.

• E009: user-reported 90%/98% frame overlay observations are qualitative, not measured box-IoUs.

• E010/S07: apparent "no reappearance" did not mean physical absence; final partially visible taxi invalidates strict target-absent coverage.

• E012: 608 should be treated as an end-of-observation/exit boundary, not a confirmed full-visibility return; annotations for ID24 end around 601.

• Not yet built: vehicle cue selection, plate handling, candidate recall measurement, reliability calibration, rejection policy and the proposed new sequential method.

Appendix C. Reference material and provenance

Primary user-supplied project documents (used as framing, not independently web-verified):

• Finalized Working Research Proposal v2, September 2026, 11 pages, uploaded as “Vehicle_Tracking_Research(6).pdf”. Key pages: 1-5 scope/research question/proposed policy, 6-8 novelty/literature, 9-10 evaluation/design.

• Vehicle Tracking + Occlusions Research Project - Phase 0 Frozen Documentation, 28 September 2026, 9 pages, uploaded as “Vehicle_Tracking_Phase_0_Frozen_Documentation(5).pdf”. Key pages: 2-3 dataset/mining workflow, 5-7 event log/scenarios, 8-9 freeze/limitations.

Phase 1 empirical source records (contents were provided in the conversation):

• Eventwise terminal logs and 12 comparison_summary.json outputs for SAM2.1 Tiny and DAM4SAM Tiny.

• Both trackers' evaluation_summary.json files and selected visual overlay frames, inspected jointly for the events discussed.

• results/phase1/phase1c_baseline_metrics.csv (24 rows), phase1c_failure_audit.csv (12 rows) and phase1c_scenario_audit.csv (10 rows), pasted as complete content.

• Manual visual-verification notes directly provided by the researcher, especially for E009 late frames, E010 presence and E001 confirmation.

The citations and original bibliography for the existing research papers remain in the research proposal. This Phase 1 execution report does not imply that the prior literature has been independently re-verified or that unreported code versions, scores and visibility annotations exist.

End-of-record control

Document cutoff: Phase 1A, 1B and 1C implementation and audits through Step 7.3, 04 October 2026. Evidence status: development/pilot. Next formal action: record Step 7.4 research priorities and complete Phase 1C Step 8 freeze separately after an explicit review.